

"PD5D midbrain single-nucleus RNA-seq hybrid selection".

Dataset Description:

This dataset contains raw FASTQ files from the midbrain single-nucleus RNA sequencing (snRNAseq) dataset with hybrid selection for the matching PMDBS samples from the PD5D cohort. The same subjects were also profiled with other omics assays including genomic DNAseq, genotyping, single-cell ATACseq, and spatial transcriptomics.

Authors:

- Kim, Kwanho; [ORCID:0000-0002-4448-3193](#); Aligning Science Across Parkinson's (ASAP) Collaborative Research Network, Chevy Chase, MD 20815, USA; Stanley Center for Psychiatric Research, Broad Institute of MIT and Harvard, Cambridge, MA 02142, USA
- Lin, Zechuan; [ORCID:0000-0003-2270-5194](#); Aligning Science Across Parkinson's (ASAP) Collaborative Research Network, Chevy Chase, MD 20815, USA; Stephen & Denise Adams Center for Parkinson's Disease Research of Yale School of Medicine, New Haven, CT 06510, USA; APDA Center for Parkinson Precision Medicine, Yale School of Medicine, New Haven, CT 06510, USA; Department of Neurology, Yale School of Medicine, New Haven, CT 06510, USA; Department of Genetics, Yale, New Haven, CT 06510, USA
- Simmons, Sean; [ORCID:0000-0002-1537-4000](#); Aligning Science Across Parkinson's (ASAP) Collaborative Research Network, Chevy Chase, MD 20815, USA; Stanley Center for Psychiatric Research, Broad Institute of MIT and Harvard, Cambridge, MA 02142, USA
- Parker, Jacob; [ORCID:0000-0001-9425-6418](#); Aligning Science Across Parkinson's (ASAP) Collaborative Research Network, Chevy Chase, MD 20815, USA; Stephen & Denise Adams Center for Parkinson's Disease Research of Yale School of Medicine, New Haven, CT 06510, USA; APDA Center for Parkinson Precision Medicine, Yale School of Medicine, New Haven, CT 06510, USA; Department of Neurology, Yale School of Medicine, New Haven, CT 06510, USA; Department of Genetics, Yale, New Haven, CT 06510, USA
- Kearney, Matthew; [ORCID:0009-0005-2106-7401](#); Aligning Science Across Parkinson's (ASAP) Collaborative Research Network, Chevy Chase, MD 20815, USA; Stephen & Denise Adams Center for Parkinson's Disease Research of Yale School of Medicine, New Haven, CT 06510, USA
- Liao, Zhixiang; [ORCID:0000-0001-7014-5465](#); Aligning Science Across Parkinson's (ASAP) Collaborative Research Network, Chevy Chase, MD 20815, USA; Precision Neurology Program of Brigham & Women's Hospital, Harvard Medical School, Boston, MA 02115, USA
- Haywood, Nathan; [ORCID:0000-0003-2213-5080](#); Aligning Science Across Parkinson's (ASAP) Collaborative Research Network, Chevy Chase, MD 20815, USA; Stanley Center for Psychiatric Research, Broad Institute of MIT and Harvard, Cambridge, MA 02142, USA
- Zhang, Jixiang; [ORCID:0000-0002-1641-8650](#); Aligning Science Across Parkinson's (ASAP) Collaborative Research Network, Chevy Chase, MD 20815, USA; Stanley Center for Psychiatric Research, Broad Institute of MIT and Harvard, Cambridge, MA 02142, USA
- Cline, Madison; [ORCID:0000-0001-9672-7738](#); Aligning Science Across Parkinson's (ASAP) Collaborative Research Network, Chevy Chase, MD 20815, USA; Banner Sun Health Research Institute, Sun City, AZ 85351, USA
- Tuncali, Idil; [ORCID:0000-0001-7437-285X](#); Aligning Science Across Parkinson's (ASAP) Collaborative Research Network, Chevy Chase, MD 20815, USA; Precision Neurology Program of Brigham & Women's Hospital, Harvard Medical School, Boston, MA 02115, USA
- Sharma, Monika; [ORCID:0000-0002-7258-1399](#); Aligning Science Across Parkinson's (ASAP) Collaborative Research Network, Chevy Chase, MD 20815, USA; Stephen & Denise Adams Center for Parkinson's Disease Research of Yale School of Medicine, New Haven, CT 06510, USA; APDA Center for Parkinson Precision Medicine, Yale School of Medicine, New Haven, CT 06510, USA; Department of Neurology, Yale School of Medicine, New Haven, CT 06510, USA
- Serrano, Geidy; [ORCID:0000-0002-9527-2011](#); Aligning Science Across Parkinson's (ASAP) Collaborative Research Network, Chevy Chase, MD 20815, USA; Banner Sun Health Research Institute, Sun City, AZ 85351, USA
- Beach, Thomas; [ORCID:0000-0003-3296-6128](#); Aligning Science Across Parkinson's (ASAP) Collaborative Research Network, Chevy Chase, MD 20815, USA; Banner Sun Health Research Institute, Sun City, AZ 85351, USA
- Dong, Xianjun; [ORCID:0000-0002-8052-9320](#); Aligning Science Across Parkinson's (ASAP) Collaborative Research Network, Chevy Chase, MD 20815, USA; Stephen & Denise Adams Center for Parkinson's Disease Research of Yale School of Medicine, New Haven, CT 06510, USA; APDA Center for Parkinson Precision Medicine, Yale School of Medicine, New Haven, CT 06510, USA; Department of Neurology, Yale School of Medicine, New Haven, CT 06510, USA; Department of Neuroscience, Yale, New Haven, CT 06510, USA
- Popic, Victoria; [ORCID:0000-0003-3181-5432](#); Broad Clinical Labs, Broad Institute of MIT and Harvard, Cambridge, MA 02142, USA
- Scherzer, Clemens; [ORCID:0000-0002-0567-9193](#); Aligning Science Across Parkinson's (ASAP) Collaborative Research Network, Chevy Chase, MD 20815, USA; Stephen & Denise Adams Center for Parkinson's Disease Research of Yale School of Medicine, New Haven, CT 06510, USA; APDA Center for Parkinson Precision Medicine, Yale School of Medicine, New Haven, CT 06510, USA; Department of Neurology, Yale School of Medicine, New Haven, CT 06510, USA; Department of Genetics, Yale, New Haven, CT 06510, USA
- Levin, Joshua; [ORCID:0000-0002-0170-3598](#); Aligning Science Across Parkinson's (ASAP) Collaborative Research Network, Chevy Chase, MD 20815, USA; Stanley Center for Psychiatric Research, Broad Institute of MIT and Harvard, Cambridge, MA 02142, USA

ASAP Team: Scherzer

Dataset Name: scherzer-pmdbs-sn-rnaseq-midbrain-hybsel, v1.0

Principal Investigator: Dr. Joshua Levin jlevin@broadinstitute.org

Dataset Submitter: Kwanho Kim kwanho@broadinstitute.org

Publication DOI: NA

Grant IDs: [ASAP-000301]

ASAP Lab: Scherzer and Levin

ASAP Project: Parkinson5D: deconstructing proximal disease mechanisms across cells, space, and progression

Project Description: Here we will develop a molecular atlas of Parkinson's disease (PD) useful for mapping GWAS/familial genetics to proximal mechanisms in five dimensions: brain cells (1D), brain space (3D), and disease stage (1D). We will reveal how genetic variants modulate transcription in specific cells in specific topographic locations of midbrain and cortex during the progression of neuropathology from healthy brains to prodromal to symptomatic disease. This research will for the first time integrate whole genome sequencing, single-nucleus transcriptomics, and high-resolution spatial transcriptomics of 100 human brains with cell-specific mechanistic analyses in model systems.

Submission Date: 2026-05-31

This dataset is made available to researchers via the ASAP CRN Cloud: cloud.parkinsonsroadmap.org. Instructions for how to request access can be found in the [User Manual](#).

This research was funded by the Aligning Science Across Parkinson's Collaborative Research Network (ASAP CRN), through the Michael J. Fox Foundation for Parkinson's Research (MJFF).

This Zenodo deposit was created by the ASAP CRN Cloud staff on behalf of the dataset authors. It provides a citable reference for a CRN Cloud Dataset